



nextwork.org

Set Up a Web App Using AWS and VS Code



sahini sourya

```
<> index.jsp ×
src > main > webapp > <> index.jsp > ...
1  <html>
2  <body>
3  <h2>Hello sourya</h2>
4  <p>This is my NextWork web application working!</p>
5  </body>
6  </html>
7
```



sahini sourya

NextWork Student

nextwork.org

Introducing Today's Project!

In this project, I will demonstrate web application that is developed from scratch and deployed using AWS cloud service and how cloud(Amazon AWS EC2 instance) where the deployment takes place to our Code from Code Editor(VS Code), I'm doing this project to learn CI/CD flow of designing and deploying the web application. This project gives you CI/CD steps from building features or changing features to deploy and available to use those features for users.

This project is part one of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project in tomorrow :)

I did this project because to kick start the 7 day DevOps challenge. Challenge was accepted now and it's great time to set up a web application in the EC2 instance.

Key tools and concepts

Services I used were VScode and Amazon EC2 and Key concepts I learnt include SSH connection using IDE, launching an instance, edit index.jsp file and using key pairs.



sahini sourya

NextWork Student

nextwork.org

Project reflection

This project took me approximately 4 hours including demo time and trouble shooting overall. The most challenging part was trouble shooting like where my EC2 instance is failing to choose AMI, EC2 permissions, stable SSH connection between terminal or VScode to EC2 Instance. It was most rewarding to see successful connection to my EC2 instance whether that's over the terminal or over the VScode SSH connection

One thing I didn't expect in this project was secret mission where we use local terminal to try edit files after using an IDE.

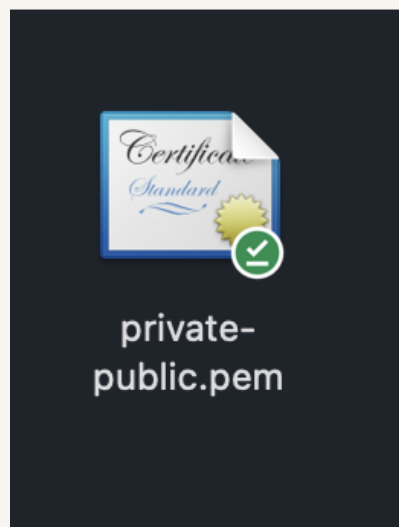


Launching an EC2 instance

What I did in this step

In this step, I will launch an EC2 instance in our AWS account and also going to set up network settings(which helps us make sure to find EC2 instance) and key pair (which helps to make sure that i have the permission to use the EC2 instance)

I started this project by launching an EC2 instance because EC2 are basically virtual computers that run on cloud, so in this project i want my web application to live entirely on cloud, so i am launching and EC2 instance to develop my application code.





I also enabled SSH

SSH is the protocol that is used by authorized users (like ourselves) to access remote servers (like EC2 instance here). It's also a type of traffic that lets us transfer data back and forth with our EC2 instance that we are connected to. I enabled SSH so that we can securely connect to my EC2 instance later and made sure that only SSH traffic was allowed from our IP address for security.

Key pairs

A key pair is a mechanism for us to get access or get connected to EC2 instances that we launch in AWS. I have created a key-pair for an EC2 instance that I am launching. A key pair works as an asymmetric key cryptography which means two keys, a public key in which AWS keeps and a private key that we download. AWS will authenticate with my private key when I want to connect to my EC2 instance later in this project by matching it to its public key.

Downloaded key pair file

Once I set up my key pair, AWS automatically downloaded the private key file called "private-public.pem". In case of my project, I should store that in my system so that I can access it later (store in a way that it can be found by me easily).



Set up VS Code

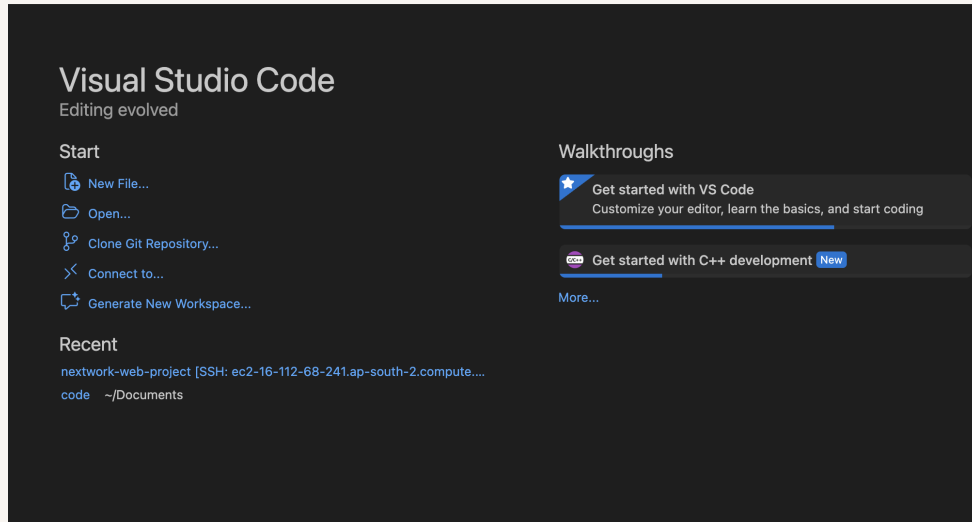
What I did in this step

In this step, I will install VS code because it its a IDE Code editor tool that was used to write software application tool, which is free of cost, i also set up a terminal and update the permission setting of .pem file that we downloaded to authenticate with our EC2 instance.

What is VS Code?

VS Code is an IDE is popular code editor that provides an Integrated Development Environment (IDE) experience.

I installed VS Code to use it to connect to your EC2 instance and work on my web application files directly.





My first terminal commands

A terminal is place for me to send commands to our computer. it's a platform that we give text intructions that we want to do in our computer. The first command I ran for this project is directing it our folder where we develop and build the code in our computer i.e., `cd ~ Desktop/Devops` which computer will go to this folder path.

Updating file permissions

I also updated my private key's permissions by using command "`chmod 400 ./private-public.pem`" in which `chmod` command generally means to change mode because it will change the permission of `.pem` file, `chmod 400` makes the file readable only by owner(us) and restrict access to everyone else.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● sahinisourya@Mac ~ % cd ~/Desktop/DevOps
● sahinisourya@Mac DevOps % ls
  DevOps-IAM-Admin_credentials.csv      private-public.pem
  newtwork-keypair.pem                  sourya-keypair.pem
● sahinisourya@Mac DevOps % chmod 400 ./private-public.pem
○ sahinisourya@Mac DevOps %
```



SSH connection to EC2 instance

What I did in this step

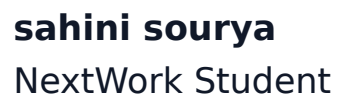
In this step, I will connect my computer(our build code) with EC2 instance(cloud) because i need to building and deploying this web application to the users through cloud, EC2 instance is our cloud computer and the code is to be in that cloud from our computer needs connection to transfer, so we connected our computer with our EC2 instance.

Connecting to EC2

To connect to my EC2 instance, I ran the command `ssh -i ~/Desktop/DevOps/private-public.pem ec2-user@ec2-16-112-68-241.ap-south-2.compute.amazonaws.com` . This command is used because ssh command means to connect securely and giving our .pem file path to connect our EC2 cloud address.

This command required an IPv4 address

A server's IPV4 DNS is like it's public address to needs to identify where the servers lives in the cloud. This steps need in our project beacuse we should tell our computer the cloud (EC2 Instance)address to launch.



nextwork.org

```
sahinisourya@Mac DevOps % ssh -i ~/Desktop/DevOps/private-public.pem ec2-user@ec2-16-112-68-241.ap-south-2.compu
te.amazonaws.com
#
##### Amazon Linux 2023
#####\
###|
#/\ https://aws.amazon.com/linux/amazon-linux-2023
V~' '~>
~~~~~
~~~~~_~_~
~/m/' ~~~
```



Maven & Java

What I did in this step

In this step, I will install Apache maven and Amazon corretto because Apache Maven provides archetypes(templates) to set up the basic structure for web application which we can use later for setting web files to create web application structure and Amazon corretto is production ready java version by AWS, as i am using Maven which operated using java language, so it requires production level java support and service versions, which is corretto.

Why I'm using Maven

Apache Maven is a tool that helps developers build and organize software projects with java. it acts as a package manager that automatically downloads external codes that project needs.

Maven is required in this project because our web application needs set up basic structure i.e., archetypes(templates) and necessary web files to create that particular structure, so these can be provided by Apache Maven.



sahini sourya

NextWork Student

nextwork.org

Why I'm using Java

Java is popular Programming Language used to build software applications varies from mobile apps to enterprise systems.

Java is required in this project because Apache Maven is used in the project which needs java to operate it, so without it, we are unable use maven to generate and build our web application



Create the Application

What I did in this step

In this step, I will generate a web application in our EC2 instance using Maven and Java because we want to generate web application that can be done by us using Maven, so i am running maven commands in our terminal for web app generation.

Creating the Java web app

I generated a Java web app using the command `mvn archetype:generate` which means we are telling maven to generate a web application using an existing template it has, it also tells maven to call the generated web application "nextwork-web-project".

Installing Remote - SSH

I installed Remote - SSH, which is a connection tool to connect our computer to another computer or server. I installed it to connect securely between my computer(VS code) to work on file and run programs in EC2 instance as if they were in my computer.



SSH configuration details

Configuration details required to set up a remote connection include host name which our address of our EC2 instance and Identity file which is location of our private key and the user that logged in our EC2 instance.

```
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/ec2-user
[INFO] Parameter: package, Value: com.nextwork.app
[INFO] Parameter: groupId, Value: com.nextwork.app
[INFO] Parameter: artifactId, Value: nextwork-web-project
[INFO] Parameter: packageName, Value: com.nextwork.app
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ec2-user/nextwork-web-project
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 17.515 s
[INFO] Finished at: 2026-01-03T15:55:23Z
[INFO] Final Memory: 20M/92M
[INFO] -----
```

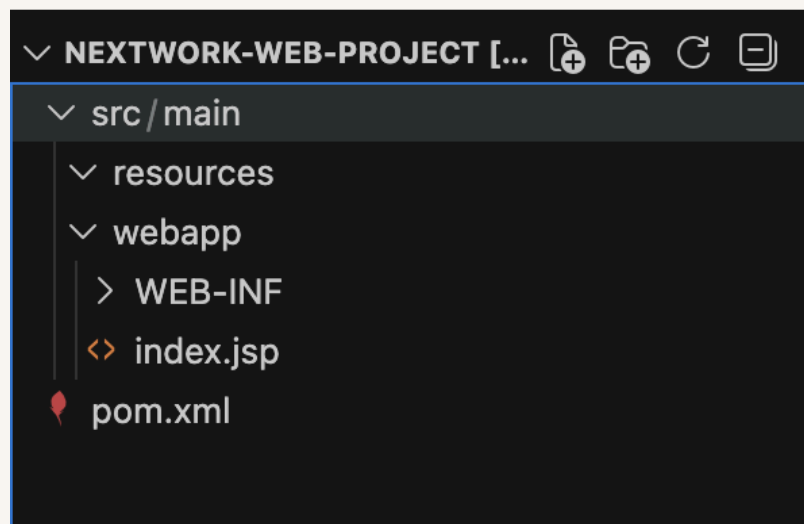


Create the Application

Exploring the project structure

Using VS Code's file explorer, I could see a bunch of folders and sub folders that define our web application. These folders organize different parts of the web application.

Two of the project folders created by Maven are `src` and `webapp`, which are `src` folder contains the information about source code files that defines how web app looks and works and `webapp` folder holds web related files like HTML, CSS and javascript etc.,.





Using Remote - SSH

What I did in this step

untill now I have connected my VScode terminal to the EC2 instance so we can give text instruction to the server but, In this step, I will connect my VS code with EC2 instance so we can now utlize VS code's IDE features like file navigation,Code editing etc., directly to EC2 instance

Updating the web app

The index.jsp is the file used in java web application that is similar to HTML file as it contains markup to display web pages and also includes java code that allows to generate dynamic content

I edited index.jsp by hello world to hello sourya(my name) and adding a paragraph tag demonstrates the working of nextwork project.



```
<> index.jsp ×  
src > main > webapp > <> index.jsp > ...  
1  <html>  
2  <body>  
3  <h2>Hello sourya</h2>  
4  <p>This is my NextWork web application working!</p>  
5  </body>  
6  </html>  
7
```



Using nano

Additional improvements

In this secret mission, I will edit the code files, in this case index.jsp using terminal instead of an IDE because it will understand us why it VScode edit and modification is easy and most likely used in production level.

Terminal vs IDE

An alternative to using IDEs is using terminal level instructions in file directory. To edit index.jsp, I ran the command `nano index.jsp` for displaying an editable window on the terminal.

Compared to using an IDE, editing index.jsp in the terminal felt little less intuitive as i have to use keyboard to navigate your cursor. I'd be more likely to use an IDE if i had lots of bits of code to update and delete, or copy or paste parts of the code.

Verifying my work

To verify my editing work in the terminal, I opened back up our VScode window that had SSH connection to my EC2 instance, It was possible to see my changes in VS Code right away because thew window shared the same EC2 instance to my local terminal.



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

GNU nano 8.3

```
<html>
<body>
<h2>Hello sourya</h2>
<p>This is my NextWork web application working!</p>
</body>
</html>
```



nextwork.org

The place to learn & showcase your skills

Check out nextwork.org for more projects

